Title: DPy: Code Smells Detection Tool for Python

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Approach and Motivation:

DPy is a tool developed to detect code smells in Python programs, addressing the gap in comprehensive smell detection tools for Python. While many existing tools focus on languages like Java, DPy aims to provide Python developers with insights into both design and implementation-level code smells. The motivation behind DPy stems from Python’s growing popularity and the need for tools that can identify maintainability issues in Python codebases. By offering detailed analysis, DPy assists developers in enhancing code quality and maintainability.

Aim and Novelty:

The primary goal of DPy is to provide a comprehensive code smell detection tool tailored for Python. Unlike existing tools that primarily target low-level syntactic issues or specific quality attributes, DPy supports detection of eight design smells and eleven implementation smells. Additionally, it offers various code quality metrics at the function, module, and class levels. The novelty of DPy lies in its ability to identify design issues at higher granularity, an area that has received limited attention in the context of Python. By exporting analysis results in CSV or JSON formats, DPy facilitates further consumption and integration into developers’ workflows.

Validation:

DPy has been validated through a comprehensive replication package that includes the tool, detailed usage instructions, and data from manual validation. The validation process involved assessing DPy’s effectiveness in detecting the supported code smells and evaluating its accuracy against known benchmarks. The results demonstrate that DPy effectively identifies both design and implementation-level code smells in Python codebases, providing valuable insights for developers aiming to improve code maintainability.